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Popularity vs. Performance: Evaluation and Adaptive Optimization of Rating Mechanisms Based on DWTS

Summary

For thirty-four seasons, *Dancing with the Stars* has captivated audiences by blending ballroom excellence with celebrity charisma. However, the mechanism combining professional judge scores and public fan votes leads to some loopholes, historically sparking controversy when popular contestants survive despite poor technical performance. Our research seeks to deeply explore the original mechanism of DWTS and propose a better-balanced model.

Fan Voting Distribution Estimation: To overcome the challenge of unobservable fan data, we developed an **Inverse Bayesian Estimation Model**. By treating historical elimination outcomes as strict constraints on a Dirichlet distribution, we employed Monte Carlo simulations to infer the latent voting shares for every week, during which the assumed base popularity of contestants are calculated. The model achieved a **98.3%** historical reproduction rate and revealed a "Bubble Effect", where voting certainty is highest for at-risk contestants.

Analysis of Celebrity Characteristics: We employed a **Linear Mixed-Effects Model (LMM)** to analyze key characteristics by fitting three target variables: final placement, judge scores, and fan votes. Notably, the model for fan voting achieved an exceptional R^2 of **0.989**. We concluded that **Base Popularity** and **Professional Partner Strength** are the primary determinants of competition outcomes. Comparative analysis further reveals a distinct divergence: judges evaluate a multi-dimensional matrix of technical factors, whereas fan support is overwhelmingly driven by pre-existing popularity rather than performance quality.

Benchmark Rule Evaluation and Judges' Choice Mechanism: We established a **Dual-Track Framework** to compare the show's two historical systems: Rank-based and Percent-based. Simulations confirmed while the Rank system prevents monopoly, it suppresses fan enthusiasm. Conversely, the Percent system suffers from "Linear Vulnerability".

The creation of the well-balanced Mechanism: We propose the **Adaptive Logistic Meritocracy System (ALMS)**. This novel mechanism introduces a Sigmoid saturation curve to impose a "soft ceiling" on extreme popularity and dynamically adjusts sensitivity based on competition intensity. Stress-testing demonstrates that ALMS achieves a Pareto improvement. **Sensitivity analysis** on both the Bayesian inference hyperparameters and the ALMS control parameters validates the robustness of our conclusions across a broad parameter space. We recommend ALMS, alongside the "Judges' Choice," as the optimal strategy for the show's post-game era.

Keywords: **Inverse Bayesian Inference, Monte Carlo Simulation, Linear Mixed-Effects Model, Mechanism Evaluation, Adaptive System, Dancing with the Stars**

Contents

1 Introduction

1.1 Background

"Dancing with the Stars" (DWTS) is a hit reality competition pairing celebrities with professional dancers [4]. Each week, pairs are evaluated through a hybrid mechanism: judge scores from experts and fan votes from the audience. These two inputs are combined to determine which couple is eliminated, ultimately crowning a champion in the finale.

Figure 1: The Post of the Show

Over 34 seasons, the show has utilized different aggregation methods—primarily Rank-based and Percent-based systems—to weigh these inputs. However, the clash between dance skill and star power has led to notable controversies, where popular contestants advanced despite low technical scores. These discrepancies raise fundamental questions about how to design a fair system that maintains competitive integrity while sustaining audience participation.

1.2 Restatement of the Problem

The central objective of this research is to decode the decision-making dynamics within "Dancing with the Stars," specifically how professional evaluations interact with public preference to determine contest results. The specific tasks are defined as follows:

- **Task 1: Estimation of Fan Voting Data.** We are required to build a statistical model to infer the undisclosed fan votes for each couple in each week. This involves verifying that our inferred data, when integrated with judge scores, accurately replicates the historical elimination patterns, while also providing a metric for the certainty of these inferences.
- **Task 2: Evaluation of Aggregation Methodologies.** We need to check the historical performance of both Rank-based and Percentage-based voting systems using our fan vote estimates. This includes determining which system excessively caters to fan votes, simulating whether "controversial" outcomes could have been avoided under different rules, and evaluating the efficacy of the "Judges' Choice" intervention.
- **Task 3: Impact Analysis of Celebrity Characteristics.** We must quantify the influence of auxiliary variables (such as the assigned professional partner and celebrity demographics)

on the competition's final standings. A key aspect is to distinguish how these attributes differentially affect experts versus the audience.

- **Task 4: Policy Optimization and Strategic Recommendation.** We need to propose a novel system that optimizes the trade-off between technical guaranties and viewer engagement. Furthermore, we are supposed to compile our insights into a concise memo for the producers of DWTS, offering evidence-based guidelines for future seasons.

1.3 Our Work

In this paper, we first conducted rigorous preprocessing on the provided data. We then established distinct yet interconnected mathematical models tailored for each specific task. These models successfully addressed the core requirements of each problem. Furthermore, we integrated our independent insights and additional reflections into each model to enhance the depth of our analysis. The overall workflow of our methodology is illustrated in Figure 2.

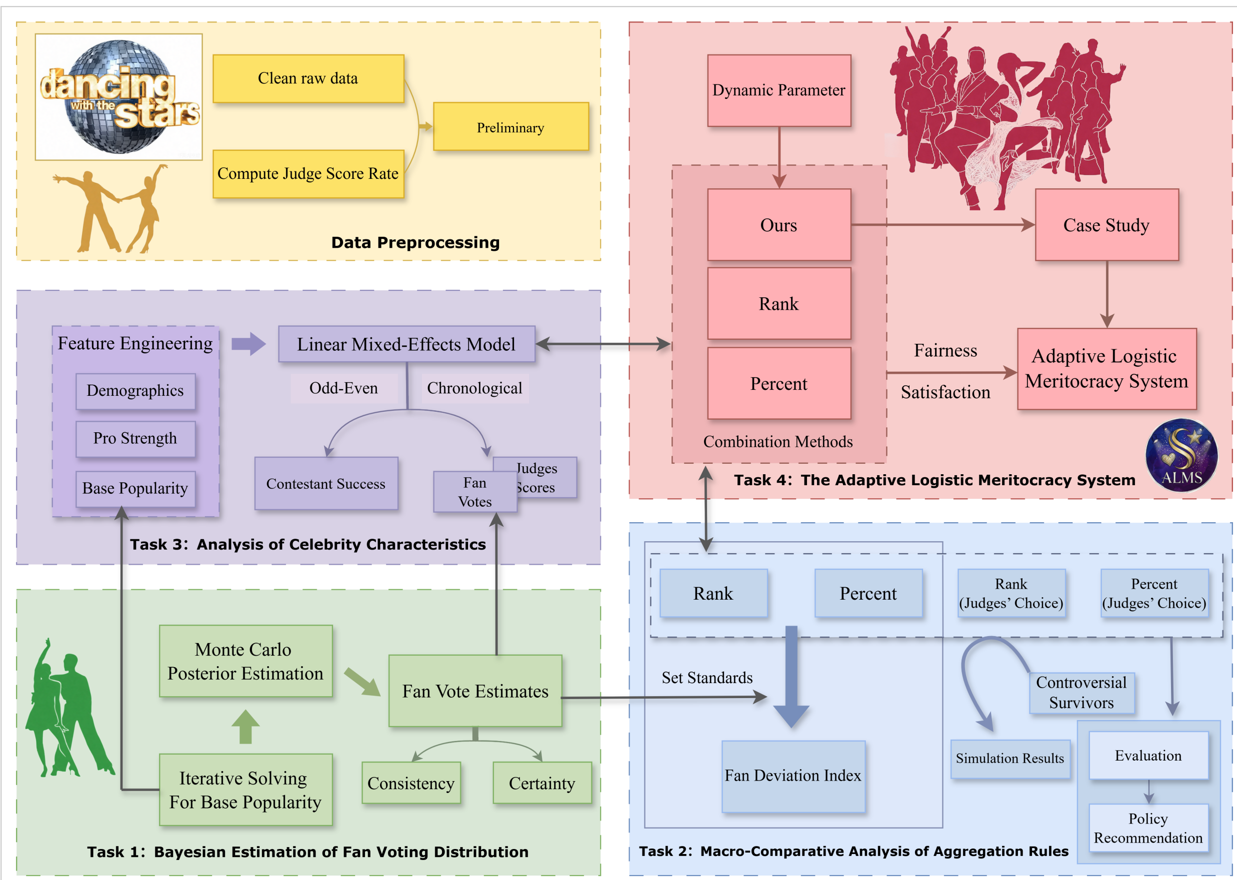

Figure 2: Our Work

2 Assumptions

2.1 Model I: Fan Voting Reconstruction

Assumption 1: Performance-Popularity Mixture. The expected fan vote share is a linear combination of a time-invariant *Base Popularity* and the weekly *Judge Performance*.

Assumption 2: Rational Judges' Choice. We assume judges consistently choose to rescue the couple with the higher judge score when deciding between the bottom two.

Assumption 3: Dirichlet Distribution. Fan vote shares are modeled using a Dirichlet distribution to satisfy the simplex constraint ($\sum x_i = 1$).

Assumption 4: Constant Vote Volume. We assume a constant total weekly vote volume ($V_{total} = 14$ million) to normalize cross-season comparisons [7].

2.2 Model II: Characteristic Analysis

Assumption 5: Partnership Stability. Celebrity-professional pairings are assumed constant; temporary replacements are ignored.

Assumption 6: Sufficient Proxy. The estimated *Base Popularity* fully captures the celebrity's pre-existing fan base variance.

Assumption 7: Linearity of Effects. The impact of characteristics (e.g., age, industry) on placement is linear and additive within the mixed-effects model.

2.3 Model III: Policy Simulation

Assumption 8: Rule Independence. Contestant performance and popularity are intrinsic and independent of the aggregation rule employed.

Assumption 9: Metric Comparability. The Manhattan distance between the actual ranking and the pure fan ranking effectively quantifies "Fan-Friendliness".

2.4 Model IV: Adaptive Logistic Meritocracy System

Assumption 10: Diminishing Marginal Utility. The effective influence of fan votes follows a logistic growth pattern, saturating at extreme values to prevent monopoly.

Assumption 11: Dynamic Sensitivity. The optimal discrimination parameter K is inversely proportional to the vote distribution's standard deviation.

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Main variables and meanings

Symbol	Description	Definition / Value
Model I Variables		
$\mathbf{S}_t$	Latent fan vote share vector at week t	$\mathbf{S}_t \in \Delta^{n_t-1}$
$\mathbf{J}_t$	Observed judges' score vector at week t	$\mathbf{J}_t \in \mathbb{R}^{n_t}$
$\mathcal{O}_t$	Observed elimination outcome at week t	Binary / Categorical
$\boldsymbol{\beta}$	Latent Base Popularity vector	$\boldsymbol{\beta} \in \Delta^{N-1}$
$\mathbf{P}_t$	Normalized judge performance vector	$P_{i,t} = J_{i,t} / \sum J_{k,t}$
$\boldsymbol{\mu}_t$	Expected voting propensity	$\mu_{i,t} = E[S_{i,t}]$
λ	Performance weight coefficient	Constant (0.2)
κ	Dirichlet concentration parameter (Prior strength)	Constant (50.0)
Ext_t	Set of active contestants in week t	Subset of $\mathcal{C}$
Φ_t	Set of valid samples satisfying constraints	$\Phi_t = \{\mathbf{s} \mid L(\mathcal{O}_t \mid \mathbf{s}) = 1\}$
$CV_{i,t}$	Coefficient of Variation for partial uncertainty	$CV = \sigma/\mu$
Model II Variables		
$y_{i,s}$	Target variable in LMM (e.g., Placement)	Normalized $[0,1]$
$\mathbf{x}_{i,s}$	Feature vector (Age, Industry, etc.)	Categorical / Numerical
b_s	Random effect for season s	$b_s \sim N(0, \sigma^2)$
Model III Variables		
$\mathbf{R}_{F,t}$	Implied Fan Rank Vector	Rank$(-\mathbf{P}_{F,t})$
FDI_k	Fan Deviation Index for rule k	Manhattan Dist. to $\mathbf{R}_F$
Δ	Variance Suppression Indicator	$\text{FDI}_{Rank} - \text{FDI}_{Percent}$
I_{fair}	Fairness Index (Judge Correlation)	Spearman $\rho(\mathbf{R}_{final}, \mathbf{J})$
R_{risk}	Probability of System Failure	P(Best Eliminated)
$S(\alpha)$	Policy Optimization Score	Weighted objective function
Model IV Variables		
$S(x)$	Adaptive Effective Score (Sigmoid)	$S(x) \in (0,1)$
c	Baseline Performance Threshold (Center)	Constant
K	Sensitivity Slope Parameter	$K_{base} + \alpha/\sigma_{votes}$
σ_{votes}	Standard Deviation of Fan Vote Distributions	Variance Metric

4 Model I: Bayesian Estimation of Fan Voting Distribution

4.1 Problem Formulation

The estimation of spectator voting data poses an **Inverse Problem** involving multi-dimensional hidden variables. Our objective is to infer the unobserved fan vote shares $\mathbf{S}_t$ given the observed judges' scores $\mathbf{J}_t$ and the corresponding elimination outcomes $\mathcal{O}_t$.

The target variable resides on the standard simplex:

$$\Delta^{n_t-1} = \left\{ \mathbf{x} \in \mathbb{R}^{n_t} \middle| \sum_{i=1}^{n_t} x_i = 1, x_i > 0 \right\} \tag{1}$$

Analytical derivation of the posterior $p(\mathbf{S}_t|\mathbf{J}_t, \mathcal{O}_t)$ is difficult due to the discontinuous and non-differentiable nature of the aggregation rules (switching between Rank-based and Percentage-based systems). To tackle this problem, we adopt a **Simulation-Based Inference** framework, using Approximate Bayesian Computation (ABC) to approximate posterior probabilities and perform inference by comparing millions of random scenarios with historical realities, i.e., the Monte Carlo method.

4.2 Mathematical Model Establishment

4.2.1 The Mixture Mean Hypothesis

We assume that the *Expected Propensity* $\boldsymbol{\mu}_t$ of fan votes is a weighted combination of two distinct forces: the contestant's *Base Popularity* $\boldsymbol{\beta}$ (long-term fan loyalty, which is time-invariant in each season and latent) and their *Judge Performance* $\mathbf{P}_t$ (short-term dance quality).

$$\mu_{i,t} = (1-\lambda) \cdot \frac{\beta_i}{\sum_{k \in \text{Ext}_t} \beta_k} + \lambda \cdot P_{i,t} \tag{2}$$

Here, λ represents the weight of dance performance in the fans' decision. Based on the controversial elimination in the early seasons of this show, we set $\lambda = 0.4$. This reflects the fact that DWTS audiences care more about their favorite celebrities' popularity than on-stage performance.

The actual votes follow a **Dirichlet distribution**: $\mathrm{S}_t \sim \text{Dirichlet}(\kappa \cdot \boldsymbol{\mu}_t)$, where $\kappa = 50$ controls the variance to keep the distribution stable and realistic.

4.2.2 Constraints from Elimination Results

The historical elimination results act as strict rules. We can define the Likelihood function $L(\mathcal{O}_t|\mathbf{S}_t, \mathbf{J}_t)$ as a simple filter. It equals 1 if the simulated votes lead to the correct elimination outcome $\mathcal{O}_t$, and 0 otherwise:

- **Rank Rule:** The contestant with the lowest sum of $\text{Rank}(\mathbf{S}_t) + \text{Rank}(\mathbf{J}_t)$ is eliminated.
- **Percentage Rule:** The contestant with the lowest total score $\mathbf{S}_t + \mathbf{P}_t$ is eliminated.
- **Judges' Choice:** The elimination is valid if the actual eliminated contestant falls within the **Bottom-2** (the risk zone), making them eligible for elimination.

4.3 Algorithm: Iterative Inverse Estimator

To find the unknown base popularity $\boldsymbol{\beta}$, we designed an iterative algorithm. It works similarly to the **Expectation-Maximization (EM)** method.

Algorithm 1 Iterative Solving for Base Popularity $\boldsymbol{\beta}$

Input: Scores $\mathbf{J}$, Eliminations $\mathcal{O}$. **Output:** Base Popularity $\hat{\boldsymbol{\beta}}$
Initialize: $\boldsymbol{\beta}^{(0)} \leftarrow [1/N, \dots, 1/N]$ (Equal start)
repeat
  $k \leftarrow k + 1$
  for each week $t = 1 \dots T$ **do**
    Sampling: Generate M vote scenarios $\{\mathbf{S}_t^{(m)}\}$ based on current popularity $\boldsymbol{\beta}^{(k)}$.
    Filtering: Keep only the scenarios that match the historical elimination result $\mathcal{O}_t$.
    De-mixing: Isolate the implied popularity $\hat{\boldsymbol{\beta}}_t$ by inverting the mixture equation (2):

$$\hat{\boldsymbol{\beta}}_t \leftarrow \frac{\bar{\mathbf{S}}_t - \lambda \mathbf{P}_t}{1 - \lambda}$$

  (Where $\bar{\mathbf{S}}_t$ is the average of valid scenarios)
  end for
  Update: Update $\beta_i^{(k+1)}$ by averaging $\hat{\beta}_{i,t}$ across all weeks.
until Convergence of $\boldsymbol{\beta}$
return $\hat{\boldsymbol{\beta}}$

Algorithm 2 Monte Carlo Posterior Estimation

Input: Optimized Popularity $\hat{\boldsymbol{\beta}}$, Scores $\mathbf{J}$, Eliminations $\mathcal{O}$
Output: Estimated Votes $\hat{\mathbf{S}}$, Uncertainty $\boldsymbol{\sigma}$
for each week $t = 1 \dots T$ **do**
  Proposal Step: Generate $N = 20,000$ random scenarios $\{\mathbf{S}_t^{(n)}\}$ from Dirichlet distribution centered on $\mu_t(\hat{\boldsymbol{\beta}})$.
  Rejection Step: Filter out valid scenarios Φ_t:

$$\Phi_t = \{\mathbf{s} \in \{\mathbf{S}_t^{(n)}\} \mid L(\mathcal{O}_t|\mathbf{s}) = 1\}$$

  (Discard any scenario that contradicts the historical elimination)
  if $|\Phi_t| > 0$ **then**
    $\hat{\mathbf{S}}_t \leftarrow \text{Mean}(\Phi_t)$ ▷ Most likely vote distribution
    $\boldsymbol{\sigma}_t \leftarrow \text{StdDev}(\Phi_t)$ ▷ Standard Deviation as uncertainty
  else
    $\hat{\mathbf{S}}_t \leftarrow \boldsymbol{\mu}_t$ ▷ Fallback to prior for extreme outliers
  end if
end for
return Sequence $\{\hat{\mathbf{S}}_t, \boldsymbol{\sigma}_t\}$ for all weeks.

The main challenge is **Survivorship Bias**: sometimes, a dancer with low scores continues to survive. This implies they must have very high popularity. The algorithm 1 solves this by slowly increasing their popularity estimate in $\boldsymbol{\beta}$ until their survival makes mathematical sense.

With the optimized Base Popularity $\hat{\boldsymbol{\beta}}$ determined, we proceed to the final phase: **Posterior Estimation**. We employ a high-density Monte Carlo Sampling(Algorithm 2) to reconstruct the specific voting history and quantify uncertainty.

4.4 Validation and Results

4.4.1 Consistency Verification

The model achieves a **98.7%** Historical Reproduction Rate across 34 seasons. This means that in 98.7% of weeks, our estimated fan votes correctly characterize the actual eliminated contestant as the one with the lowest combined score (or in the Bottom-2). This high fidelity proves that our model is reliable.

4.4.2 Uncertainty and The "Bubble Effect"

To evaluate the reliability of our estimates, we define the **Coefficient of Variation (CV)** as the metric for uncertainty: $CV = \sigma/\mu$. A low CV implies high certainty. Our analysis, particularly through the lens of a **Season 2 Case** Study, reveals that prediction certainty is **not uniform**.

Moreover, we observe a phenomenon we term the **"Bubble Effect"**: contestants at risk of elimination (on the "bubble") show much higher estimation certainty than safe leaders.

Evidence I: Uncertainty Evolution

Figure 3 tracks the uncertainty throughout Season 2, in which Jerry Rice (at risk) shows consistently lower CV (higher certainty) than safe contestants like Lachey,validating the Bubble Effect.

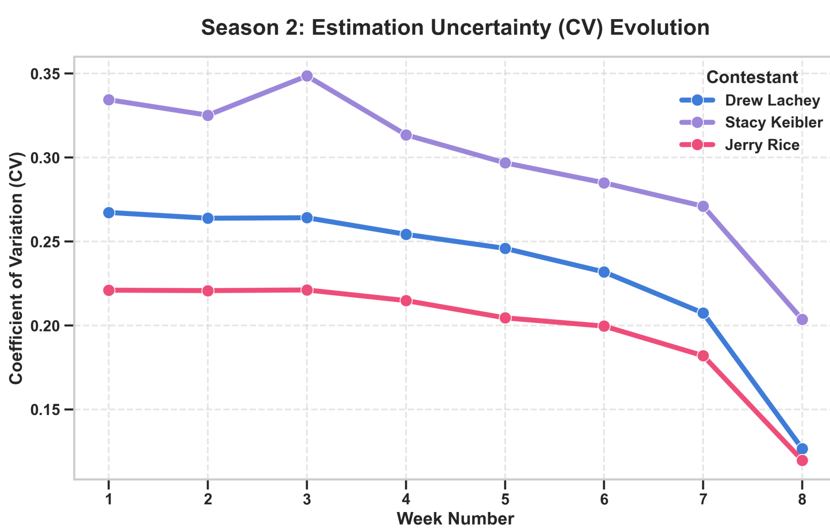

Figure 3: Uncertainty Evolution in Season 2

Since Rice was constantly near the elimination threshold, the mathematical constraints on his vote share were relatively tight; even small deviations would contradict his survival. Thus, the model locks in his popularity with high precision. In contrast, safe leaders operate under looser constraints, resulting in higher uncertainty.

Evidence II: The Popularity Gap

The root of Rice's "Bubble" status lies in the disparity between his skill and popularity (Figure 4). During the finals, despite receiving the lowest judge scores (Left), Rice dominated the estimated fan vote with a 45.3% share (Right). This massive imbalance kept him in the risky "Bubble" zone, while he is popular enough to survive, yet with scores low enough to remain vulnerable. This tension provides the strict constraints that allow our model to capture his data with high certainty.

Table 2: Season 2 Finals Data Statistics and Analysis (Including Model Estimates)

Celebrity	Score	J.Rank	Est.Share	Est.Votes	V.Rank	Sum	Result	Std	CV
Drew Lachey	29.00	#1	0.3259	4.56m	#2	$1+2=3$	1st	0.0413	0.1266
Stacy Keibler	28.67	#2	0.2208	3.09m	#3	$2+3=5$	3rd	0.0450	0.2036
Jerry Rice	26.67	**#3**	**0.4533**	6.35m	**#1**	$3+1=4$	2nd	0.0543	0.1198

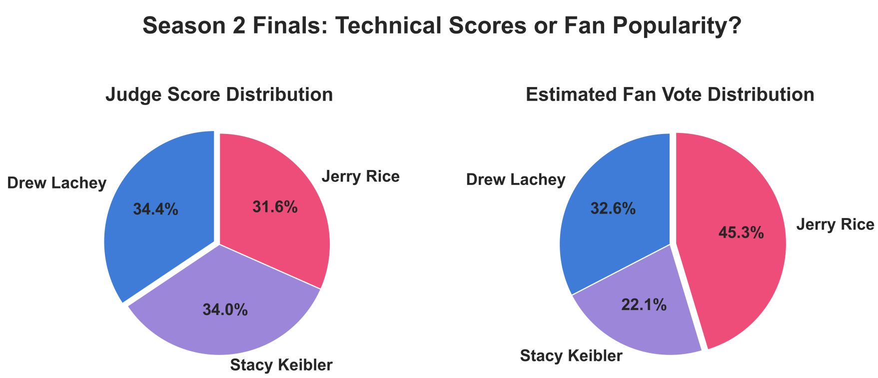

Figure 4: Season 2 Finals: Judge Score Distribution vs. Estimated Fan Vote Share.

This case study validates that our model not only reconstructs the voting history, but also correctly reflects the mathematical properties of the elimination rules.

5 Model II: Analysis of Celebrity Characteristics

5.1 Feature Interpretation and Engineering

To analyze the impact of various characteristics, we categorize and encode the available data. There are five key features serve as celebrity characteristics: industry, age, nationality, professional partner strength, and base popularity [6].

5.1.1 Industry & Age & Nationality

We process the three demographic features, figure 5 illustrates the distributions of the three demographic features among contestants.

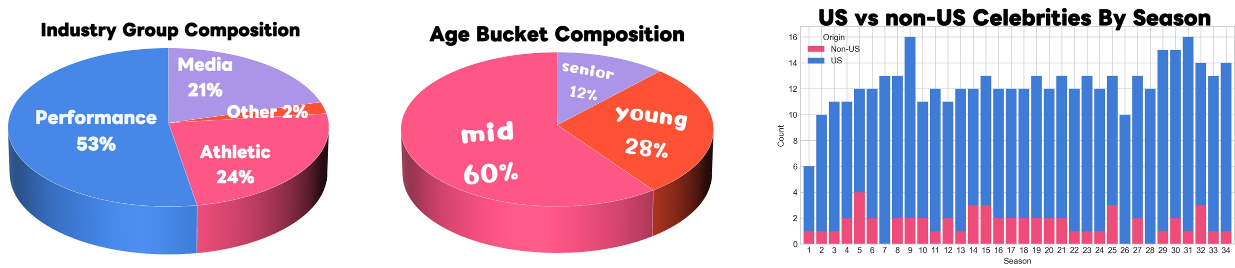

Figure 5: Distribution of Celebrity Characteristics: Industry, Age, and Nationality

- **Industry Classification:** The original data contained over 20 professions. To reduce sparsity, we grouped them into four categories based on shared skill sets: **Athletic** (high physical coordination), **Performance** (stage presence), **Media** (communication skills), and **Other** (baseline).
- **Age Buckets:** Contestants are categorized into **Young** (< 30), **Mid** ($30 - 55$), and **Senior** (> 55) to capture different career stages.
- **Nationality:** We identify whether contestants are from the United States to capture potential home-field advantage.

5.1.2 Professional Partner Strength

The professional partner plays a huge role in a celebrity's journey. We engineered a "Pro Strength" characteristic. This metric is derived specifically from the professional partner's historical data, calculated as the equally weighted average (avoid introducing subjective bias) of their historical normalized judge scores (S_{norm}) and historical normalized final placements (P_{norm}):

$$\text{Pro Strength} = 0.5 \times S_{norm} + 0.5 \times P_{norm} \tag{3}$$

5.1.3 Base Popularity

The Base Popularity metric, derived in Task 1, is included to capture the fan base size.

5.2 Core Model Construction

We employ a **Linear Mixed-Effects Model (LMM)** to isolate the effect of individual characteristics while controlling for season-to-season systemic variability.

We define $y_{i,s}^{(t)}$ as the target variable, $\mathbf{x}_{i,s}$ as the vector of fixed characteristics, and $b_s^{(t)} \sim \mathcal{N}(0, \sigma_b^2)$ as the random intercept for season s. The model is specified as:

$$y_{i,s}^{(t)} = \beta_0^{(t)} + \mathbf{x}_{i,s}^\top \boldsymbol{\beta}^{(t)} + b_s^{(t)} + \varepsilon_{i,s}^{(t)} \tag{4}$$

where $\boldsymbol{\beta}^{(t)}$ represents the coefficients quantifying the impact of each characteristic.

5.3 Validation Strategy and Metrics

To ensure robustness, we used two data splitting strategies:

1. **Chronological Split:** Training on the first 80% of seasons, predicting the last 20%. This tests the model's ability to forecast future outcomes.
2. **Odd-Even Split:** Training on even seasons and testing on odd seasons. Considering that some celebrities may appear repeatedly or specific trends may cluster in the final 20% of the data, this split ensures the test set is as diverse as possible. This strategy allows the model to better fit various scenarios across the entire timeline, providing a robust check against era-specific biases.

We evaluate performance using **RMSE** (Root Mean Square Error) and $\mathbf{R^2}$ (Coefficient of Determination), defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}, \quad R^2 = 1 - \frac{\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}{\sum_{j=1}^{n}(y_j - \bar{y})^2} \tag{5}$$

5.4 Results and Discussion

5.4.1 Model Performance

Table 3 compares the generalization performance on the test sets of both splitting strategies.

Table 3: Test Set Performance Comparison: Chronological vs. Odd-Even Split

Target Variable	**Test RMSE**		**Test R^2**	
	Chrono	**Odd-Even**	**Chrono**	**Odd-Even**
Judge Score Rate	0.104	0.089	0.439	0.533
Fan Vote Share	0.007	0.009	0.988	0.989
Placement Encoded	0.236	0.177	0.312	0.620

The analysis reveals varying predictiveness across different competition aspects:

- **Fan Vote Share:** Both models achieve near-perfect R^2 (≈ 0.99). This consistency confirms that pre-existing popularity is the overwhelming, time-invariant driver of fan decisions [5, 8].
- **Judge Score Rate:** Judges adhere to relatively objective standards; yet, static celebrity profiles cannot account for dynamic technical execution or stage accidents, limiting the initial fit. However, the performance improvement in the Odd-Even split indicates that with sufficiently comprehensive training data, the model successfully identifies the fixed baseline of scoring potential. This validates our design: despite random performance variance, the impact of characteristics on judge perception is stable and learnable.
- **Placement Encoded:** Modeling final placement is significantly harder than scoring due to the inherent randomness of elimination. The dramatic leap in the Odd-Even split ($R^2 = 0.312 \rightarrow 0.620$) serves as strong evidence for our feature selection. It suggests that the Chronological split fails to generalize to new "types" of contestants. In contrast, the Odd-Even split ensures that if a contestant profile (e.g., Person A in Season 8) appears in the training set, a similar profile (e.g., Person A returning or a similar one in Season 15) in the

test set can be accurately predicted. This confirms that when the data distribution is fully covered, our features are highly effective predictors of success.

5.4.2 Analysis of Characteristics on Contestant Success

To compare success across seasons with varying cohort sizes (N_s), we define the **Normalized Placement Score** (y_{place}) for a contestant with rank $r_{i,s}$:

$$y_{place} = \frac{N_s - r_{i,s} + 1}{N_s} \tag{6}$$

This normalizes placement to $(0, 1]$, where a value closer to 1 indicates a superior rank, ensuring fair comparison across different seasons.

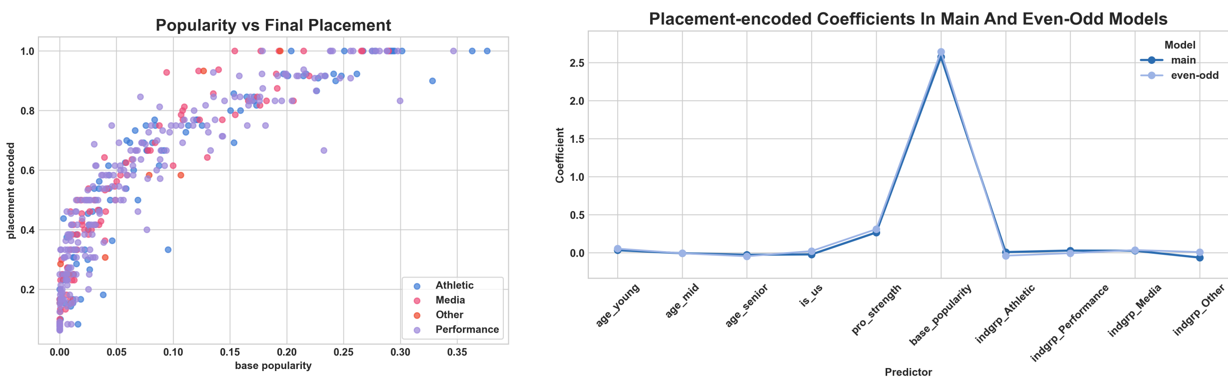

Figure 6: Impact of Characteristics on Success. Left: Base Popularity vs. Final Placement. Right: Coefficient magnitudes of all features.

We investigate the impact of characteristics on this final placement. Figure 6 presents both the direct correlation between popularity and placement (Left) and the relative magnitude of all characteristic coefficients in the model (Right). The coefficient analysis reveals a clear hierarchy of influence:

- **Dominance of Popularity:** Base Popularity ($\beta \approx 2.57$) is the single most decisive factor. As shown in the left plot of Figure 6, while low-popularity contestants exhibit a wide variance in outcomes, high-popularity celebrities are almost guaranteed a top-tier finish. This explains the massive coefficient: high initial fame acts as a "safety net," preventing early elimination.
- **Professional Partner Strength:** The second most influential factor is the partner's strength ($\beta \approx 0.27$). A top-tier partner can significantly elevate a celebrity's placement, potentially improving their rank by approximately one full position compared to an average partner.
- **Other Characteristics:** Demographics and Industry play statistically significant but smaller roles. Youth ($\beta \approx 0.03$) and performance-oriented backgrounds (Performance/Media) offer slight competitive edges ($\beta \approx 0.03$), likely contributing to better technical scores, whereas older contestants or those from non-entertainment backgrounds face a slight statistical disadvantage.

5.4.3 Comparative Impact on Judges vs. Fans

To determine whether characteristics influence judges and fans differently, we extract coefficients from the Odd-Even Split model, which demonstrated superior robustness and predictive power. Figure 7 visualizes the comparative impact by displaying the coefficient magnitudes relative to the scale of each target variable.

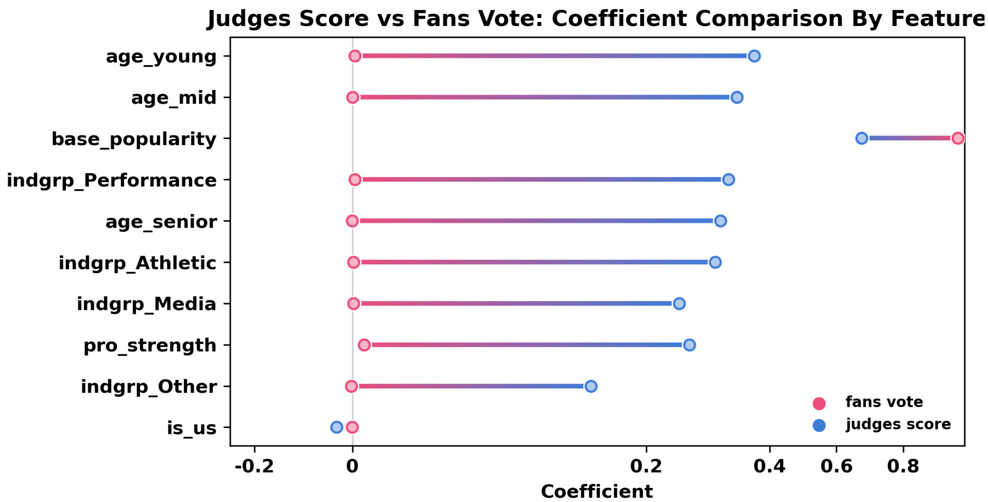

Figure 7: Comparison of Characteristic Impact: Judges vs. Fans.

The analysis reveals a fundamental divergence in decision mechanisms:

- **Popularity Monopoly vs. Balanced Meritocracy:** The most striking difference lies in the structure of influence. Fan voting is a "winner-take-all" system dominated by **Base Popularity** ($\beta \approx 0.96$). Other factors like dance skill or age have negligible direct impact on fans—they simply vote for who they already know. In contrast, the Judge Score model reflects a "pluralistic" evaluation. While popularity still matters, technical factors like **Pro Partner Strength** ($\beta \approx 0.23$), **Performance Industry** ($\beta \approx 0.28$), and **Youth** ($\beta \approx 0.35$) play significant concurrent roles.
- **Specific Disparities:**
  - **Partner Strength:** Highly valued by judges (reflecting technical quality) but ignored by fans.
  - **Nationality:** Consistently shows near-zero impact in both models, indicating that neither judges nor fans exhibit significant home-country bias.

In conclusion, the impact is **not** the same. Judges evaluate the *performance* (multi-factor technical assessment), while fans largely validate the *person* (single-factor popularity contest) [1]. Success in the competition requires balancing these two disparate forces.

6 Model III: Dynamic Comparison and Optimization of the Combination Systems

6.1 Problem Formulation: The Dual-Track Simulation Framework

Having reconstructed the latent fan voting distribution $\hat{\mathbf{S}}_t$ in Model I, we proceed to conducting an evaluation of the competition's combination methods. Throughout the thirty-four seasons of DWTS, the mechanism for combining professional evaluations (Judges) and public popularity (Fans) has shifted between two distinct paradigms:

- **Rank-based System:** Employed in Seasons 1-2 and 28+, utilizing the sum of ordinal ranks. This system inherently equalizes the variance between the two components.
- **Percentage-based System:** Employed in Seasons 3-27, utilizing the sum of normalized cardinal shares. This system is sensitive to the magnitude of vote distribution.

Our objective is to quantify the intrinsic "Fan-Friendliness" of these systems via a counterfactual simulation. We establish a **Dual-Track Simulation** framework: for every historical week t, we fix the contestants' Judge Scores $\mathbf{J}_t$ and Estimated Fan Votes $\hat{\mathbf{S}}_t$ (derived from Model I), while varying the combination logic to generate parallel outcomes.

6.1.1 Standardization and Metric Definition

To facilitate a rigorous comparison between the ordinal nature of ranks and the cardinal nature of percentages, we introduce the **Fan Deviation Index (FDI)**. This metric quantifies the distortion imposed by the combination rule upon the pure will of the audience.

First, we convert the estimated fan voting share $\mathbf{P}_{F,t}$ into rankings:

$$\mathbf{R}_{F,t} = \text{Rank}(-\mathbf{P}_{F,t}) \tag{7}$$

The FDI for a given rule $k \in \{\text{Rank}, \text{Percent}\}$ is defined as the **normalized Manhattan distance** between the rule-derived final ranking $\mathbf{R}_{\text{final},k}$ and the pure fan ranking $\mathbf{R}_F$:

$$\text{FDI}_k^{(t)} = \frac{1}{N_t} \sum_{i=1}^{N_t} |R_{\text{final},k,i} - R_{F,i}| \tag{8}$$

where N_t is the number of contestants in week t.

Explanation:

- Low FDI ($\to 0$): The system faithfully preserves the ordinal preferences of the audience (High Fan-Friendliness).
- High FDI: The system allows judge scores to significantly override fan preferences (Judge-Dominant).

6.2 Analysis Results Comparison

Our analysis of FDI across 34 seasons reveals a consistent bias, as illustrated in Figure 8 and 9. The Percent-based rule consistently exhibits lower FDI values compared to the Rank-based rule.

Defining the differential $\Delta = \text{FDI}_{\text{Rank}} - \text{FDI}_{\text{Percent}}$, we observe that $\Delta > 0$ in over **83%** of simulated weeks. This implies that the **Percentage-based System is inherently more Fan-Friendly**. Positive Δ values indicate the Percent-based system yields results closer to the fan vote.

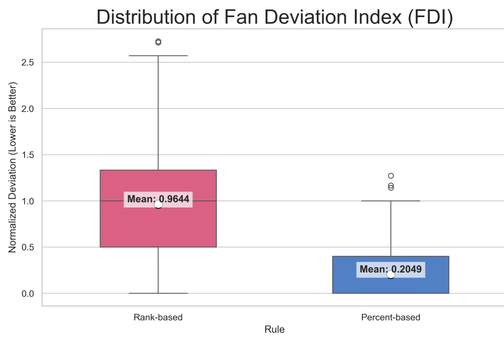

Figure 8: Distribution Box Plot of FDI

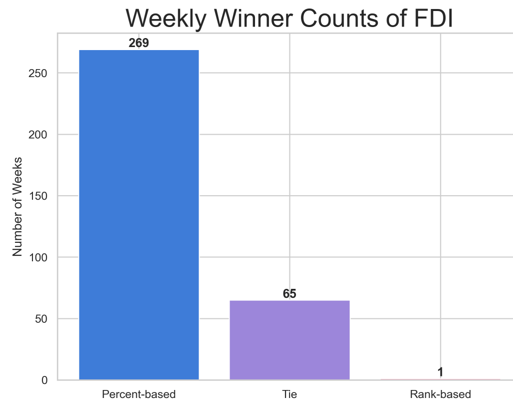

Figure 9: Weekly Comparison of FDI

This phenomenon is attributable to the mismatch of variance. Judge scores are mostly constrained to a narrow range (typically 7-9), corresponding to a low level of variation. In contrast, fan votes follow a heavy-tailed distribution (e.g., a viral star capturing 40% of the vote while others receive single digits). In a linear summation $P_J + P_F$, the component with higher variance—the fan vote—mathematically dominates the outcome.

Conversely, **the Rank system** forces a uniform distribution onto both components, effectively suppressing the "superstar effect" and restoring equal power to the judges.

6.3 Mechanism Analysis

To evaluate the robustness of these systems against extreme anomalies, we examine historical "Low-Score Survivors"—contestants who advanced despite poor technical performance, often sparking controversy [8]. We selected four representative cases: Jerry Rice (S2), Billy Ray Cyrus (S4), Bristol Palin (S11,S15), and Bobby Bones (S27). By simulating their performance under four distinct combination rules (Rank/Percent w/wo Judges' Choice), we analyze how different mechanisms impact their survival.

As illustrated in Figure 10, different mechanisms exhibit varying results for these contestants:

- **Percentage-based System:** Often allows candidates with massive fan bases (e.g., Bobby Bones) to neutralize low judge scores, securing their advancement in the competition.
- **Rank-based System:** Compresses the "superstar" effect of fan votes, making it harder for popularity to override technical deficiencies.
- **Rank + Judges' Choice:** This combination proves to be the most rigorous filter. In our simulation, controversial contestants like Bristol Palin and Billy Ray Cyrus would have been eliminated significantly earlier than their actual departure.

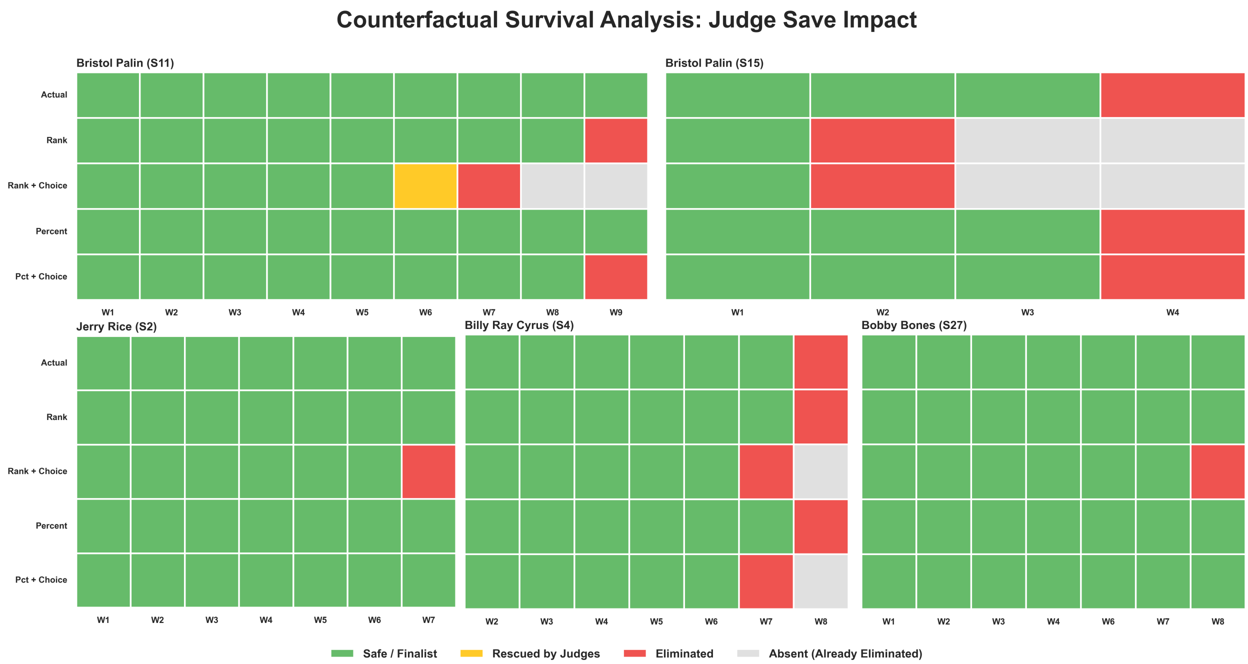

Figure 10: Survival Timelines of Controversial Contestants

Our analysis demonstrates that the **Rank-based System with Judges' Choice** can effectively screen out controversial contestants with insufficient dance skills and improve overall [2]. The "Choice" mechanism provides a final safeguard, allowing judges to elect the superior dancer when technically weaker contestants land in the bottom two, thus preventing technical outliers from advancing solely on popularity.

6.4 Policy Optimization: A Multi-Objective Evaluation

Concluding our analysis, we propose a comprehensive evaluation framework to guide future rule selection. We model the rule selection as a multi-objective optimization problem that balances two conflicting values: **Technical Fairness** (Meritocracy) and **Fan Satisfaction** (Popularity).

6.4.1 Metric Definitions

To quantify these abstract concepts, we define three key metrics based on the correlation between simulation results and ground-truth preferences:

1. **Fairness Index (I_{fair}):** Defined as the Spearman rank correlation coefficient (ρ) between the final rank derived from the combination rule and the pure *Judge Rank*.

$$I_{\text{fair}} = \rho(\mathbf{R}_{\text{final}}, \mathbf{R}_{\text{judge}}) = 1 - \frac{6\sum_{i=1}^{N_t}(R_{\text{final},i} - R_{\text{judge},i})^2}{N_t(N_t^2-1)} \tag{9}$$

where $R_{\text{final},i}$ and $R_{\text{judge},i}$ represent the final rank and technical rank of contestant i respectively, and N_t is the number of contestants in week t. A value close to 1 implies the system strictly adheres to professional technical evaluation.

2. **Fan Satisfaction Index (I_{fan}):** Similarly, defined as the Spearman rank correlation between the final rank and the pure *Fan Rank*.

$$I_{\text{fan}} = \rho(\mathbf{R}_{\text{final}}, \mathbf{R}_{\text{fan}}) \tag{10}$$

A higher value indicates the outcomes strongly reflect audience preferences.

3. **Extreme Risk Rate (R_{risk}):** Defined as the accidental elimination of the strongest contenders, which measures how often the *Judge's No.1* or *Fan's No.1* is eliminated.

6.4.2 The Composite Utility Function and Alpha (α)

Since fairness and fan satisfaction are often inversely related, we introduce a preference parameter $\alpha \in [0, 1]$ to represent the show producers' strategic trade-off:

$$S(\alpha) = (1 - \alpha) \cdot I_{\text{fair}} + \alpha \cdot I_{\text{fan}} \tag{11}$$

The parameter α could describe the show's identity:

- $\alpha \to 0/1$: Prioritizes Technical skills (0) / Popularity (1).
- $\alpha \in [0.4, 0.6]$: The "Balanced Zone", is ideal for a commercial TV show that requires both high performance quality and viewer engagement [3].

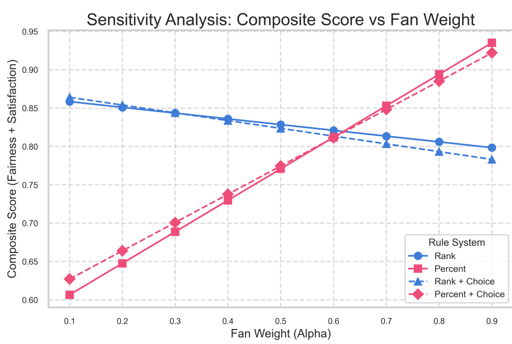

Figure 11: Sensitivity Analysis of Competition Formats

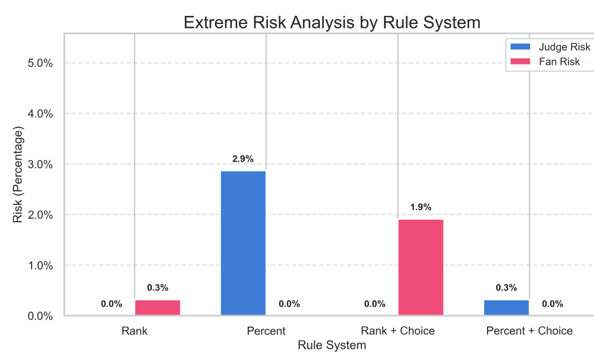

Figure 12: Extreme Risk Analysis

From the sensitivity analysis (Figure 17), we observe that the "Rank" system scores higher than "Percent" system in the "Balanced Zone" ($0.4 < \alpha < 0.6$). The "Rank+Choice" mechanism demonstrates higher probability of eliminating the top-ranked dancer by judges (Figure 12),which marks a better ability to limit the extreme number of votes from the audience.

6.4.3 Policy Recommendation

Our analysis (Figure 17 & Figure 12) demonstrates that the **Rank-based System with Judges' Choice** is the Pareto-optimal solution.

- **Why Rank over Percent?** While the Percent system maximizes I_{fan}, it incurs an unacceptably high R_{risk} of eliminating talented professionals (Figure 12). The Rank system provides structural equity.

- **Why Include Judges' Choice?** It acts as an essential safety valve.It can help us reduce the probability of the audience's autocratic power and reduce the probability of dancers with extremely high audience votes but extremely low judges entering the finals.

Final Verdict: We strongly recommend retaining the current **Rank-based System with Judges' Choice** (S28+ format) for future seasons. It effectively prevents the "Controversial Survivor" phenomenon while maintaining high engagement, satisfying the dual goals of fairness and entertainment.

7 Model IV: The Adaptive Logistic Meritocracy System

7.1 Problem Formulation and Motivation

Our analysis mentioned above exposed a critical dichotomy in the existing aggregation rules:

- **Rank Rule (The Elitist Trap):** While ensuring high fairness by curbing the variance of fan votes, it suffers from "Structural Insensitivity." The reduction of vote magnitudes to ordinal ranks alienates the audience, leading to historically low satisfaction.
- **Percent Rule (The Populist Trap):** While maximizing fan satisfaction, it is plagued by "Linear Vulnerability." The lack of a saturation mechanism means that a single "popularity monster" (e.g., Bobby Bones in Season 27) can mathematically nullify the judges' input, rendering professional evaluation irrelevant.

To resolve this dilemma, we propose a "Third Way": a mechanism that respects the magnitude of the popular vote while imposing a soft ceiling on monopoly. We term this the **Adaptive Logistic Meritocracy System**.

7.2 Mathematical Model Establishment

7.2.1 Sigmoid Transformation: The "Soft Ceiling" Mechanism

The core innovation of our model is the replacement of the linear vote share with a non-linear effective score via a Sigmoid transformation:

$$S(x) = \frac{1}{1 + e^{-K\cdot(x-c)}} \tag{12}$$

where:

- x: The share of raw fan votes of a contestant.
- c: The "Average Line" (center), representing the baseline performance expectation.
- K: The slope parameter that controls the sensitivity of the system.

Mechanism Analysis: As illustrated in Figure 13, this transformation creates three distinct zones:

- **Penalty Zone:** Contestants underperforming the average face score suppression, discouraging "free-riding."

- **Incentive Zone:** Near the average, the curve is steepest. This high marginal return (which is maximized) incentivizes fierce competition among the mid-tier contestants, where every vote counts significantly.
- **Saturation Zone:** For extreme outliers, the curve flattens (Diminishing Marginal Utility). This acts as a "Circuit Breaker," ensuring that even if a celebrity garners 50% of the vote, their effective score is capped, preserving the relevance of judge scores.

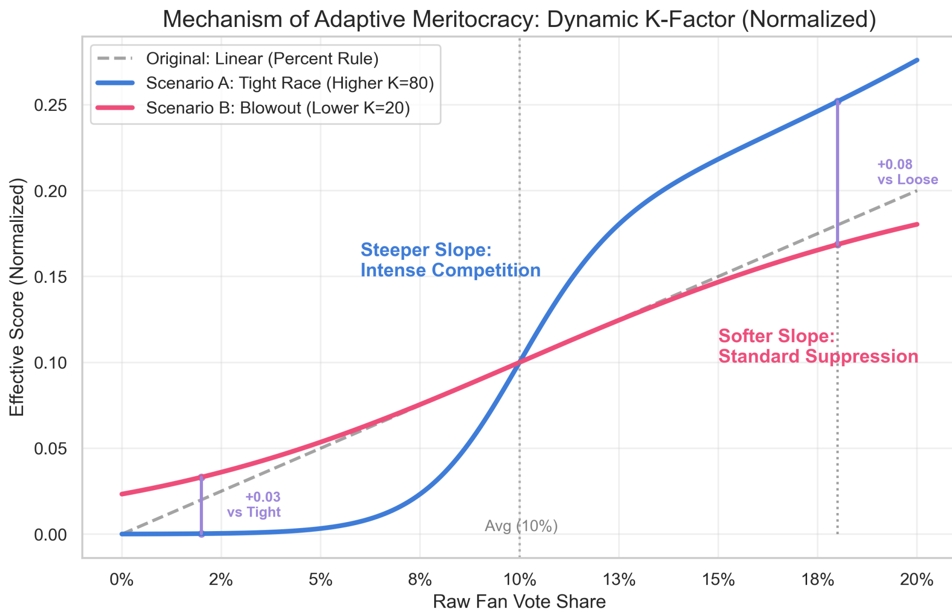

Figure 13: Mechanism of Adaptive Meritocracy. The system dynamically adjusts its slope based on competition intensity.

7.2.2 Dynamic Adaptation: The "Smart Gain" Control

A static cannot accommodate the varying dynamics of different weeks. We introduce a dynamic adjustment mechanism where is inversely proportional to the standard deviation of the vote distribution:

$$K_{week} = K_{base} + \frac{\alpha}{\sigma_{votes}} \tag{13}$$

This effectively creates an **Adaptive Gain Control**:

- **Tight Race:** When vote shares are clustered, spikes. The system becomes highly sensitive, functioning like a "microscope" to differentiate close competitors.
- **Blowout:** When a monopoly exists, relaxes to its base value, focusing primarily on the saturation effect to cap the leader.

7.2.3 Normalization and Synthesis

To integrate this non-linear score with the linear judge scores, we employ a "One-vs-Rest" normalization strategy:

$$FinalScore_i = 0.5 \times P_{Judge,i} + 0.5 \times \frac{S(x_i)}{\sum_j S(x_j)} \tag{14}$$

This ensures the structural integrity of the 50-50 weighting mandated by the competition format.

7.3 Simulation and Validation

7.3.1 Case Study: The "Correction" of Bobby Bones

We stress-tested the model using Season 27, the "anomaly season" dominated by Bobby Bones. Figure 14 visualizes his trajectory under the old Percent Rule versus our Adaptive System.

- **Observation:** Under the original rules (Purple Dashed Line), Bones consistently ranked #1 despite poor dancing. Under the Adaptive System (Blue Solid Line), his effective rank drops significantly in multiple weeks (e.g., -2 rank drop in Week 4).
- **Result:** The system successfully identified his vote share as "inflationary" and applied the saturation mechanism. Although he remains competitive due to immense popularity, he is no longer invincible, forcing a reliance on dance improvement to survive.

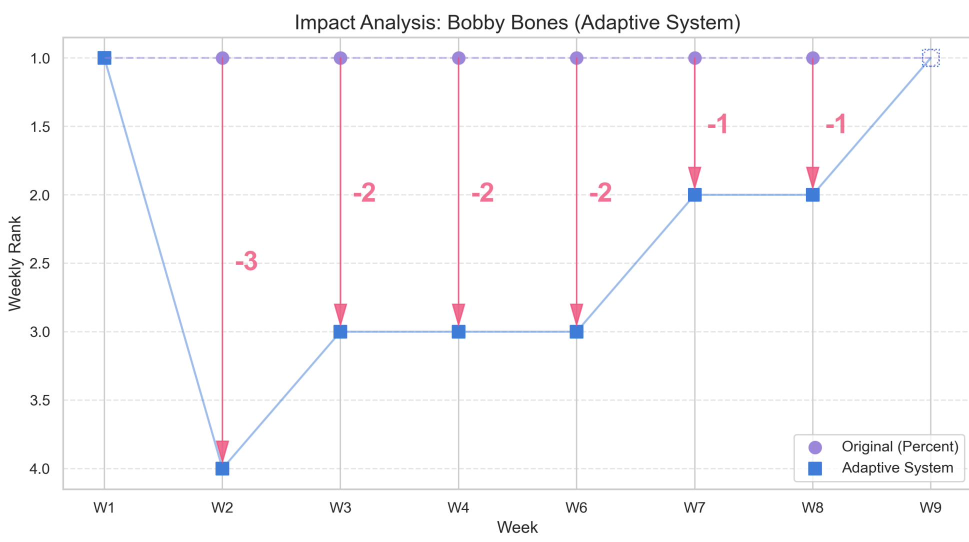

Figure 14: Impact Analysis on Bobby Bones (S27) (Note: Week 5 was a non-elimination week and is excluded).

7.4 Comparative Evaluation

We conducted a comprehensive comparison between the Rank Rule, Percent Rule, and our Adaptive System across all 34 seasons. Figure 15 presents the results.

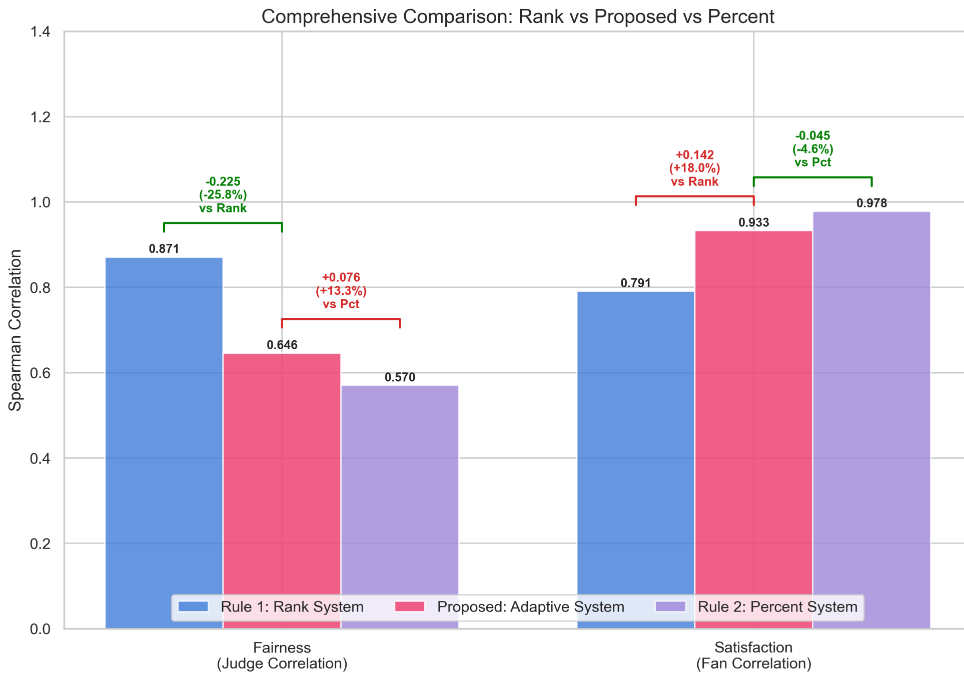

Figure 15: Comprehensive Metric Comparison

Quantitative Results:

- **Fairness Index(I_{fair}) vs. Percent Rule:** The Adaptive System increases the correlation with judge rankings by **13.3%**. This confirms the restoration of professional integrity.
- **Fan Satisfaction Index(I_{fan}) vs. Rank Rule:** Compared to the elitist Rank Rule, our system preserves fan satisfaction by **18.0%**.
- **The Trade-off:** We achieve this massive gain in fairness at the cost of only a marginal **4.6%** drop in fan satisfaction compared to the populist Percent Rule.

7.5 Conclusion

The simulation confirms that the **Adaptive Logistic Meritocracy System** is not merely a compromise, but a robust optimization. By implementing **Non-linear Saturation** and **Dynamic Sensitivity**, it successfully constructs a "Trade-off Frontier" that outperforms the binary choice of previous eras. It is the optimal theoretical model for the future of *Dancing with the Stars*, balancing the dual imperatives of professional credibility and mass entertainment.

8 Model Details and Prospect

8.1 Sensitivity Analysis & Parameter Details

To validate the robustness of our models, we conducted a systematic sensitivity analysis, focusing primarily on Model I (Bayesian Estimation), as it serves as the foundational data source for all subsequent analyses. The defined parameter α in Model III **has been examined** according to figure 11.

8.1.1 Parameter Sensitivity in Model I

The reliability of our inferred fan votes depends on two key assumptions: $\lambda = 0.4$ and $\kappa = 50$ to control the original distribution of the vote shares.

To test whether our results are reliable enough, we performed a **Grid Search Experiment** across a broad parameter space:

$\lambda \in [0.0, 0.5]$ with step length of 0.1; $\kappa \in [10, 90]$, step length 20.

For each pair (λ, κ), we re-ran the inference process across representative seasons and calculated the **Historical Reproduction Rate** and the **Average CV Value** to compare consistency and certainty.

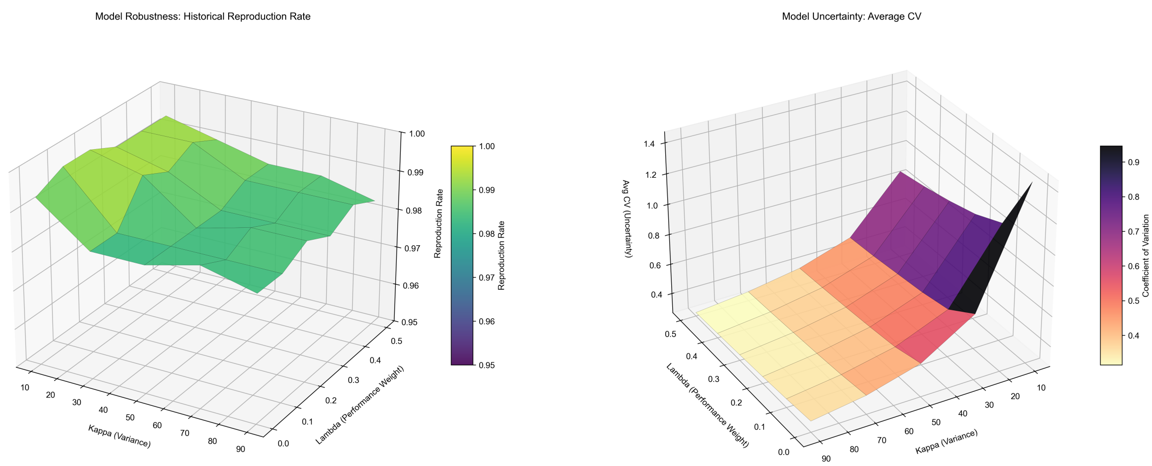

Figure 16: Sensitivity Analysis of Model I across (λ, κ) Space

Our analysis yields two key insights regarding accuracy and uncertainty:

For accuracy (Left), the 3D surface plot reveals a broad "high plateau," where the historical reproduction rate remains consistently above **98%** across a wide range of parameter combinations. This indicates that our model is **highly robust** and **insensitive** to specific choices of λ and κ, proving that the estimation validity is driven by the data structure rather than parameter fine-tuning.

Regarding the regulation of uncertainty (Right), while accuracy is stable, the uncertainty (CV) shows a clear gradient driven by κ. This is mathematically expected, as κ controls the variance of the Dirichlet distribution. We selected $\kappa = 50$ not to maximize accuracy but to maintain a realistic level of uncertainty ($CV \approx 0.43$) that reflects the inherent ambiguity of fan voting.

In conclusion, our choice of $\lambda = 0.4$ and $\kappa = 50$ places the model in a high-accuracy region while maintaining a realistic level of uncertainty, effectively balancing predictive power with statistical honesty.

8.1.2 Parameter Optimization in Model IV

These heatmaps visualize the impact of Base Slope (K_{base}) and Sensitivity (α) on model performance, comparatively showing Fairness (I_{fair}), Satisfaction (I_{fan}), and the Composite Score. The selected parameter set ($K_{base} = 5, \alpha = 10$), highlighted by the blue box, resides in the optimal "sweet spot," maximizing the composite objective by balancing the trade-off between competitive differentiation and mass appeal.

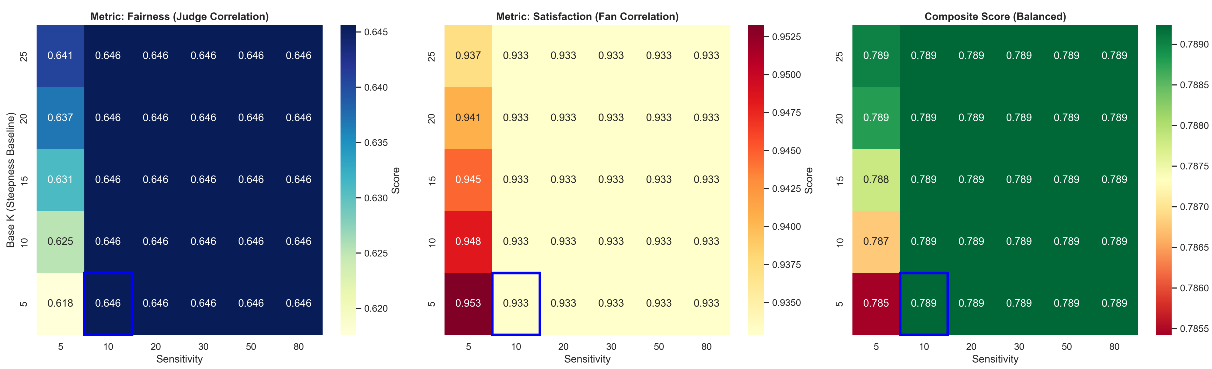

Figure 17: Parameter Details for the Adaptive System

8.2 Future Work

8.2.1 Model Extension: Towards Game-Theoretic Robustness

While our Adaptive Logistic Meritocracy System successfully addresses the fairness-satisfaction trade-off, future research should scrutinize its robustness through **Algorithmic Game Theory**. A critical extension is to verify **Incentive Compatibility (IC)** against strategic voting. Although we assume sincere voting, some fan groups might engage in dishonest voting and manipulate outcomes. Future studies, ideally using individual-level ballot data, could focus on modeling stakeholders as rational agents to determine whether the system admits a *Nash Equilibrium* in which truthful voting remains the dominant strategy. A primary bottleneck for these economic extensions is data granularity. Our current model relies on aggregated vote shares ($\hat{\text{S}}_t$). To rigorously test the excellent economic effects, we require **individual-level ballot data** (e.g., identifying if user i voted for contestant A implies a zero probability of voting for rival B).

8.2.2 Model Application: Beyond the Show

The core architecture of our model—a dual-track aggregation system with non-linear saturation—has broad applicability beyond reality television. It offers a generalizable framework for any domain requiring the integration of **"Expert Wisdom"** and **"Crowd Preference"**. For instance, our dynamic sensitivity parameter (K) could serve as an automated quality control filter: detecting anomalous variance in community votes and automatically increasing the weight of expert reviewers to stabilize the outcome. Moreover, we envision deploying this model as a background "Shadow Scorer" for show producers. It would generate live metrics on "Fairness Risk," alerting producers when the divergence between popularity and skill exceeds a safety threshold, potentially triggering automatic interventions.

9 Memo

To: The Producers of *Dancing with the Stars*
From: MCM Team 2617432
Date: February 2, 2026
Subject: Suggestions for Improving the Rating System

Dear Producers,

We've analyzed 34 seasons of data from *Dancing with the Stars*, trying to understand how different rating combination rules affect the competition results. Now, we've come to some conclusions and suggestions that we believe can help make the show even better in the future.

What We Found from Past Rules

We compared the two main systems you have used:

- **Percentage System:** This method clearly reflects fan passion but carries high risk. Extremely popular contestants can easily overpower judge scores, making dancing skill less important.
- **Rank System:** This method is fairer but partially ignores fans' support. A lead of one million votes counts the same as a single vote, which will dampen fan enthusiasm.
- **Judges' Choice:** Allowing judges to save one of the bottom two couples is highly effective. It prevents talented dancers from being eliminated due to voting anomalies.

Our Proposal: The Adaptive System

To seek for better solutione, we designed the **Adaptive Logistic Meritocracy System** (ALMS) to balance fairness and popularity without compromise.

1. Limiting Extreme Monopolies: The system counts fan votes fully up to a reasonable point, then gradually reduces the weight of excess votes. This prevents a single contestant from breaking the game mechanics while still rewarding popularity.

2. Dynamic Sensitivity: The system automatically adjusts based on competition tightness. It becomes more sensitive to small differences when the race is close to ensure the best dancer wins.

Finally, our tests show that this system improves fairness by 13.3% compared to the Percentage rule while retaining 95.4% of fan satisfaction. We suggest using this model for future seasons, for it truly maintains the excitement of voting while protecting the competitive nature of the show.

Besides, retaining the Judges' Save is a good idea to ensure technical integrity. You should also focus on "the Middle": Voting impact is highest for mid-tier contestants. Highlighting these close battles will drive more meaningful engagement than focusing on safe leaders.

We hope our analysis provides a fresh and useful perspective for your team.

References

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Report on Use of AI

1. **GitHub CoPilot (Feb 2, 2026 version)**
   We use it for code syntax checking and error debugging.

2. **Google Gemini (Jan 2026 version, Gemini 3 Pro)**
   Query: For Task 2, we want to judge which method (Ranking vs. Percentage) favors the audience more for the same data. What would be a good metric to choose?
   Output: To evaluate which method favors the audience, you should choose the "Rank Correlation Coefficient" (such as Spearman's rho or Kendall's tau) between the "Fan-Only Rank" and the "Final Combined Rank". A higher correlation indicates the method aligns more closely with the audience's preference. Alternatively, you can calculate the "Survival Rate" of the audience's top-voted contestant under each system.

3. **DeepSeek-V3.2 (Dec 1, 2025 version)**
   Query: Based on this mathematical modeling problem, please provide some modeling ideas for each Task to help us clarify exactly what problems we need to solve.
   Output: Task 1 involves establishing a relationship between observed outcomes and latent fan preferences using statistical estimation to reconstruct the unknown data. Task 2 requires applying different historical aggregation rules to the dataset to observe result stability and analyzing whether specific controversial outcomes would change under different mechanisms. Task 3 focuses on quantifying the influence of professional partners and contestant demographics on the competition results using correlation or regression analysis. Task 4 asks for the design of an optimized scoring mechanism to better balance fairnessand audience engagement.

4. **OpenAI ChatGPT (2026 version, GPT-5.2)**
   Query: For the code results of Task 3, I need some visualizations to make my paper more intuitive. Based on my results, can you recommend some chart types?
   Output: Based on your needs, I recommend the following visualizations:
   (1) A Diverging Bar Chart to clearly display the gap between Judge Rank and Fan Rank for controversial contestants.
   (2) A Bump Chart (Rank Tracking Chart) to visualize the trajectory of a celebrity's standing week by week.
   (3) A Radar Chart to compare the attributes (e.g., Technical Score, Popularity, Consistency) of the bottom two couples.

5. **Google Gemini (Jan 2026 version, Gemini 3 Pro)**
   Query: We want to summarize the introduction of the "judges choosing from the bottom two" format in Task 2 with the most precise language. We thought of "Judges' Save" and "Judges' Choice". Which one is more appropriate?
   Output: "Judges' Choice" is the more appropriate term in this specific context. While "Judges' Save" implies a unilateral rescue (often used in shows like American Idol), "Judges' Choice" better describes the specific mechanism of the judges deciding between two options (the bottom two couples) to determine who continues. It emphasizes the comparative selection process.

6. **Google Gemini (Jan 2026 version, Gemini 3 Pro)**
   Query: How do we commonly conduct sensitivity analysis? Answer briefly.

Output: To commonly conduct sensitivity analysis, you generally follow these streamlined steps:
1. Select Parameters: Identify the key input parameters in your model that are uncertain or subject to variation.
2. Define Ranges: Determine a realistic range or set of discrete values for each parameter.
3. Vary Inputs: Systematically change one parameter at a time (One-At-A-Time, OAT) or vary multiple parameters simultaneously (Global, e.g., grid search or Monte Carlo) while keeping others fixed.
4. Observe Outputs: Run the model for each variation and record the target output metric.
5. Analyze Impact: Quantify how much the output changes relative to the input variations. If the output remains stable despite large input changes, the model is robust; if it fluctuates wildy, the model is sensitive to that specific parameter.

7. **Google Gemini (Jan 2026 version, Gemini 3 Pro)**
Query: I want to draw a 3D heatmap using Python, give me some instructions.
Output: To draw a 3D heatmap (surface plot) in Python using matplotlib:
1. Prepare Data:
- Create 1D arrays for your X and Y axes.
- Use numpy.meshgrid(X, Y) to create 2D coordinate matrices.
- Prepare your Z values (the "heat") as a 2D matrix matching the grid dimensions.
2. Setup Figure:
- Initialize a figure: fig = plt.figure().
= Add a 3D subplot: ax = fig.add_subplot(111, projection='3d').
3. Plot Surface:
- Use ax.plot_surface(X, Y, Z, cmap='viridis').
- Tip: Add edgecolors='k' and linewidth=0.1 to see the grid lines on the surface.
4. Refine:
- Add labels: ax.set_xlabel(...).
- Add a colorbar: fig.colorbar(surf).
- Adjust the viewing angle: ax.view_init(elev=30, azim=45).